

Group 24's Wildfire Prediction Model

Project Proposal

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Introduction:

Wildfire ignitions and spread are driven by increasingly volatile, non-linear interactions across atmospheric conditions, vegetation fuel moisture, and terrain topography. Because microclimatic variables like relative humidity and wind speed fluctuate rapidly over short temporal and spatial scales, predicting localized wildfire ignition risk remains an inherently complex challenge.

Rather than relying solely on static or broad spatial risk estimates, this project will explore the feasibility of capturing dynamic environmental interactions using modern data science workflows. To achieve this, we propose incorporating high-resolution ERA5 weather reanalysis, MODIS satellite vegetation indices, and USGS topographic data with USDA historical fire records to construct a machine learning pipeline that models daily 24-hour wildfire ignition probabilities across California grid cells. Our primary objective is to build and evaluate a machine learning model capable of predicting potential daily wildfire ignitions across Californian spatial grids using these multi-modal environmental datasets.

Motivation:

The core motivation behind the wildfire prediction project stems from a critical intersection of humanitarian urgency and technical skill cultivation. Globally, accelerated climate change has intensified the frequency and devastation of wildfires - threatening human lives, destroying ecosystems, and causing economic damage. By developing predictive machine learning models, this project aims to provide early, actionable risk indicators that could assist communities and emergency responders. Predicting wildfire risk in advance directly supports decision-making related to evacuation planning, strategic resource allocation, and emergency readiness.

From an educational perspective, this project provides an opportunity to apply advanced data science concepts to a real world, high-impact environmental problem. The workflow will include collecting, cleaning, and analyzing wildfire and environmental data, and identifying factors associated with wildfire occurrence. Machine learning techniques will then be used to develop and evaluate predictive models. Through this process, the project aims to strengthen practical skills in data processing, exploratory data analysis, feature selection, model development, and performance evaluation while demonstrating how data science can be applied to address environmental challenges.

Evaluation:

The study area will involve partitioning California into 5km x 5km spatial grid cells and training our models on historical environmental observations. With each grid cell and day representing an observation labeled according to whether a wildfire ignition occurred. Historical observations will be divided into training, validation, and testing sets while preserving temporal separation where possible to reduce data leakage and better evaluate the model's ability to predict future wildfire events.

Wildfire ignitions are expected to represent a relatively small portion of all grid cells observations, so primary evaluation metrics will include wildfire-class precision, recall, F1-score, and Precision-Recall Area Under Curve (PR-AUC). Receiver Operating Characteristic Area Under Curve (ROC-AUC) and overall accuracy reported as supplementary metrics as they cannot be relied upon independently.

Recall will be particularly important because failing to identify actual high-risk wildfire conditions or events (a false negative) carries a significantly higher real-world cost than a false alarm. Thus, precision will be considered to ensure that improving recall does not result in excessive false alarms. As a result, F1 score and Precision-Recall Area Under Curve will also provide measures to evaluate the tradeoff between these outcomes.

Multiple classification models will be trained under the same evaluation framework and compared using these metrics. The final analysis will also examine model errors and feature importance to identify environmental conditions associated with correct predictions, false positives, and false negatives.

Resources:

Datasets -

- **USDA Forest Service (FPA-FOD):** Historical wildfire occurrence data (1992 - present) providing spatial ignition coordinates, discovery dates, and final fire size attributes for positive label generation.
- **ECMWF ERA5 Reanalysis:** Gridded hourly and daily atmospheric variables (2m air temperature, surface relative humidity, 10m wind speeds, surface pressure, and total precipitation) at ~ 31km spatial resolution.
- **NASA MODIS (Terra/Aqua):** Moderate Resolution Imaging Spectroradiometer vegetation products (MOD13A2 / MYD13A2), providing 16-day composite Normalized Difference Vegetation Index (NDVI) at 1km spatial resolution to track live fuel moisture and drought dynamics.
- **USGS Digital Elevation Model (DEM):** High-resolution terrain surface elevation data used to derive spatial topographic features, including slope gradient and aspect facing angles.

Software and Libraries -

- **Language Environment:** Python 3.x
- **Geospatial & Data Processing:** GeoPandas, Rasterio, Xarray, Shapely, PyProj, Pandas, NumPy
- **Machine Learning Frameworks:** Scikit-Learn, XGBoost, LightGBM, PyTorch
- **Visualization & Geospatial Mapping:** Matplotlib, Seaborn, Folium / Leafmap

Hardware and Compute Environment -

- **Primary Platform:** Standard free Google Colab notebook instances for shared code execution and cloud-based model training.
- **Local Development:** macOS (Apple Silicon M3 and Intel) and Windows laptop environment for initial code development, data exploration, and script testing.
- **Storage and Data Ingestion:** Google Drive for shared file storage and the Google Earth Engine Python API (geemap) for spatial raster extraction and grid alignment.

Contributions:

The project responsibilities are divided among team members according to core project phases. While individual members lead specific sections, all team members will maintain continuous communication and collaborate across each stage to ensure seamless task handovers and integration. Each assigned lead acts as the primary manager for their phase rather than working entirely in isolation.

The division of primary roles is as follows:

- **Data Acquisition and Preprocessing:** Aditya Pundhir
- **Model Selection, Architecture, and Training:** Amsal Moiz
- **Model Evaluation, Testing, and Result Visualization:** Bob Bobanza Monga

For written deliverables and final reporting, each member will author the sections corresponding to their lead technical responsibilities. In addition, all members will jointly contribute to analyzing overall project performance, interpreting final results, and writing the final discussion.